



Presented by :
Kaustubh Bodade , Co-founder

RAKSHAVAHAN.●

Detect. Decide.
Defend.



RakshaVahan

Today's Agenda

- 01** Introduction
- 02** Problem Statement
- 03** Our Innovative Solutions
- 04** Discover Our Services
- 05** Size of Market
- 06** Key Competitive Advantages
- 07** Accomplishments Date
- 08** Use of Funds for MVP only
- 09** Meet our Mentors & Advisors
- 10** Meet our Team

Introduction



SOLUTION

A radar system for the
detection of explosives
from moving vehicles

RAKSHAVAHAN is an AI-driven defence technology platform built to protect moving convoys from explosive threats such as roadside IEDs and vehicle-borne bombs. Our mission is to build India's first intelligent convoy protection system RADAR — combining radar, spectroscopy, and AI to detect and respond to threats before they strike.



RakshaVahan

Problem Statement



● PROBLEM

Defense and security convoys remain critically exposed while ***in motion***— current explosive detection systems are static, limited to checkpoints, and ineffective once vehicles are on the move.

Incidents like the **Pulwama attack** reveal a massive technology gap: **no real-time detection or automated protection exists for moving vehicles against roadside, attached, or approaching explosives.**

This leaves thousands of defense and VIP convoys vulnerable every day?

Sources : https://timesofindia.indiatimes.com/india/pulwama-attack-allowing-civilian-vehicles-on-route-proved-disastrous/articleshow/68000622.cms?utm_source=chatgpt.com



RakshaVahan

Our Innovative Solutions

RAKSHAVAHAN unites Radar, GPR, Raman, NQR and IMS sensors into one AI-powered defence radar – a smart fusion system that detects, identifies, and prevents explosive threats in real time, protecting India's convoys, soldiers, and critical assets



Radar



GPR



Raman



NQR



IMS

Solution 01



A universal, vehicle-mounted Radar Module that uses GPR , NQR , Standoff Raman Spectrometer & IMS enables real-time detection and rapid blast protection for moving vehicles and can be made for vehicle using retrofitments.

Solution 02



RAKSHAVAHAN fuses radar, spectroscopy, and AI to deliver the world's first fully-mobile explosive detection and defence system.

Solution 03



RAKSHAVAHAN unites Radar, GPR, Raman, NQR, and IMS sensors into one AI-powered defence radar — a smart fusion system that detects, identifies, and prevents explosive threats in real time, protecting India's convoys, soldiers, and critical assets

Discover Our Services

01

System Integration & Deployment

Custom installation and field setup for defence and security fleets.

02

Real-Time Command Dashboard

Live radar view, GPS-tracked alerts, and AI analytics.

03

AI Analytics & Mission Intelligence

Smart data-driven threat heatmaps and predictive insights.



RakshaVahan

Size of Market

India's defence and homeland security ecosystem is undergoing rapid modernization, with an emphasis on mobility, AI integration, and indigenous protection technologies.

Today, more than 150,000 defence and paramilitary vehicles operate without any active, on-the-move explosive detection or defence system — creating a massive untapped opportunity.

With the Government of India investing heavily in Make in India, iDEX, and Atmanirbhar Bharat, solutions like RAKSHAVAHAN — which enhance convoy survivability and reduce import dependency — are positioned to become the standard protection layer across multiple domains.

₹3,500+ Cr

Total Addressable Market: ₹3,500+ Cr in India + \$500M Global Export Potential.

6.5%
CAGR

RAKSHAVAHAN Business Model



- Hardware Sales / Modules**
Sale of detection modules (Radar + Sensor Pod)
One-time sale or lease
- SaaS Platform (AI & Analytics)**
Cloud-based dashboard for threat analytics
Annual subscription
- AMC & Maintenance Contracts**
Annual service, calibration & field upgrades
10-15% of system cost / year
- OEM Partnerships / Licensing**
Integration with vehicle OEMs (Tata, Mahindra)
Royalty per vehicle / module

Target Market – Securing India's Most Critical Convoys

	Market Value (₹ Crores)
Primary Market – Defence & Paramilitary Forces 10,000+ high-risk military and paramilitary transport vehicles including Indian Army convoy trucks, and CRPF/BSF troop carriers	3,500+ TAM
Secondary Market – Homeland & Civil Security State Police, NSG/SPG VIP transport, explosive ordnance disposal units, and border security forces	SAM
Tertiary Market – Civilian & High-Value Convoy Applications	Serviceable Available Market

Key Competitive Advantages

Feature / Parameter	RAKSHAVAHAN (India)	Counter-IED Systems (USA)	Vehicle Protection Systems (EU / Israel)	Static Scanning Systems (India)
Detection Type	Radar + Multi-Sensor Fusion (GPR, NQR, Raman, IMS)	Ground Penetrating + IR + Static Radar	Millimeter Wave + EO + LIDAR	Stationary X-ray / Scanning
Mobility	✔ Fully Mobile (operates while convoy moves)	✘ Stationary or semi-mobile	✘ Limited to checkpoint or static	✘ Fixed infrastructure
Threat Coverage	Buried, Attached, Chemical & Concealed	Buried IEDs only	Concealed weapons & metal objects	Limited to static cargo
Response System	Automated alert + optional shield deploy	Manual / delayed response	Alert only	Manual scanning
Integration Flexibility	Retrofit & OEM integration	Military OEM-only	Limited OEM scope	Fixed installations
AI & Data Fusion	Real-time AI analytics + predictive learning	Manual or semi-automated	Basic data fusion	None
Cost Advantage	💰 ~20% of imported system cost	Very High (>\$2M/unit)	High (>\$1M/unit)	₹1Cr+ per site
End-User Target	Defence, Paramilitary, VIP, Civil Security,International market	Military-only	Government contracts	Civil / Customs use



RakshaVahan

Accomplishments Date



SEPT 2025

**END OF
2028**

**AFTER FIELDS
TEST OF MVP**

Problem Identification & Defence Validation.

Completed stakeholder interviews (Army, CRPF), Pulwama risk analysis, technology feasibility verified

Prototype Development .

Working prototype under assembly with radar, sensor fusion software, and vehicle integration. Test readiness at DRDO Proving Grounds, performance validation (Pd, FAR, latency), data collection.

Certification & Compliance & OEM Partnership & Scale.

Secure DRDO, DGQA, and lab safety certifications for radar emissions and field use
Partnerships with Tata Defence, Mahindra, Bharat Forge for manufacturing and OEM-fitment.



RakshaVahan

Meet Our Mentors and Advisors



PROF.(DR.)RAJESH M. BODADE

SENIOR PROFESSOR,

*MILITARY COLLEGE OF
TELECOMMUNICATION ENGINEERING,
MHOW*



BRIGADIER DEEPAK PURI
COMMANDER , FCTS

**MILITARY COLLEGE OF
TELECOMMUNICATION
ENGINEERING , MHOW**

Guided by senior Indian Defence officers with decades of field experience and leadership in national security technology programs.



RakshaVahan

Meet Our Team

A focused and mission-driven team united by a shared goal — strengthening India's defence capabilities through innovation and excellence.



KAUSTUBH BODADE
CO-FOUNDER



AYUSH AHIRE
CO-FOUNDER



RakshaVahan

Presented by :
Kaustubh Bodade , Co-founder

Thank You

for your time
and attention



+91 9422141054



founder@rakshavahan.in
rakshavahantech@rakshavahan.in



Proposed Address to be Registered:
Military College Of Telecommunication
Engineering , MHOW